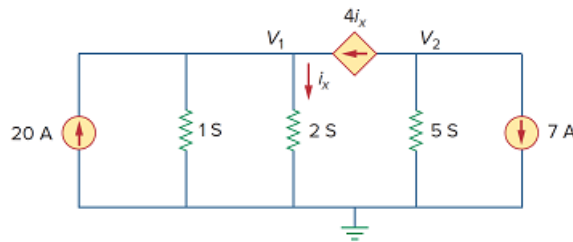


Problem

Write the node-voltage equations by inspection and then determine values of V_1 and V_2 in the circuit of Fig. 3.114.

Figure 3.114:



Step-by-step solution

Step 1 of 3

Refer to the circuit diagram in Figure 3.114 in the textbook.

Apply Kirchhoff's current law at node v_1 .

$$20 + 4i_x = v_1 + 2v_1$$

From the Figure 3.114, the current i_x is

$$i_x = 2v_1$$

Therefore,

$$20 + 4(2v_1) = 3v_1$$

$$20 + 8v_1 = 3v_1$$

$$5v_1 = -20 \dots\dots (1)$$

Step 2 of 3

Apply Kirchhoff's current law at node v_2 .

$$5v_2 + 4i_x + 7 = 0$$

Substitute $2v_1$ for i_x .

$$5v_2 + 4(2v_1) + 7 = 0$$

$$5v_2 + 8v_1 = -7 \dots\dots (2)$$

Therefore, the node voltage equations are

$$\begin{cases} 5v_1 = -20 \\ 8v_1 + 5v_2 = -7 \end{cases}$$

Step 3 of 3

From equation (1),

$$5v_1 = -20$$

$$v_1 = -\frac{20}{5}$$

$$= -4 \text{ V}$$

Therefore, the voltage, v_1 is $\boxed{-4 \text{ V}}$.

Substitute -4 V for v_1 in equation (2).

$$5v_2 + 8(-4) + 7 = 0$$

$$5v_2 - 32 + 7 = 0$$

$$5v_2 = 25$$

$$v_2 = 5 \text{ V}$$

Therefore, the voltage, v_2 is $\boxed{5 \text{ V}}$.